

## WIPRO--JAVA SCRIPT-ASSIGNMENT-→

### 1.Lab Question: Alert, Confirm, and Prompt – Quiz Game

Source code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Quiz Game</title>

<script>

    function startQuiz() {

        alert("🎯 Welcome to the Quiz Game!");

        let name = prompt("What's your name?");

        if (!name || name.trim() === "") {

            alert("❌ You didn't enter your name. Exiting game.");

            return;

        }

        let ready = confirm(`Hi ${name}! Are you ready for the quiz?`);

        if (!ready) {

            alert("Okay, maybe next time!");

            return;

        }

        let answer = prompt(

            "How many states are there in India?\n" +

            "a) 28\n" +
```

```
"b) 29\n" +  
"c) 26"  
);
```

```
if (!answer) {  
    alert("You didn't answer. Game over!");  
    return;  
}
```

```
answer = answer.trim().toLowerCase();  
if (answer === "a" || answer === "28") {  
    alert(`✅ Correct! Well done, ${name} 🎉`);  
} else {  
    alert("❌ Oops! The correct answer is 28.");  
}  
}
```

```
</script>
```

```
</head>
```

```
<body onload="startQuiz()">
```

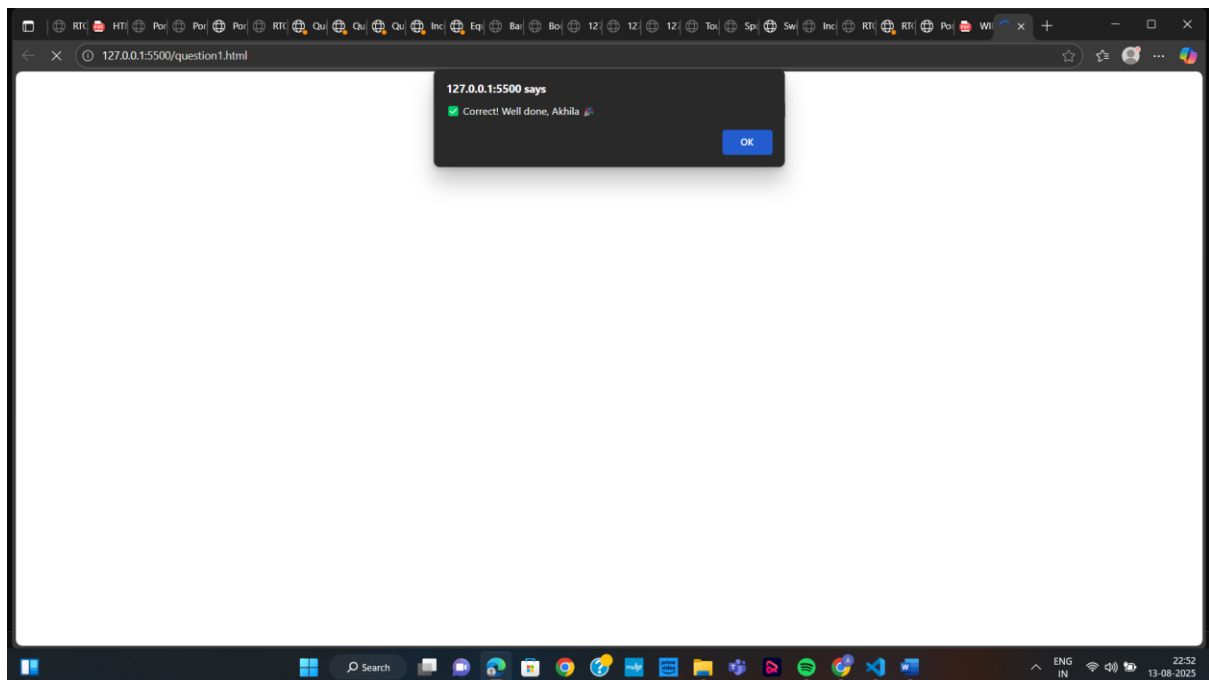
```
<h1>Simple Quiz Game</h1>
```

```
<p>Reload the page to play again.</p>
```

```
</body>
```

```
</html>
```

Output:



## 2.Lab Question: Income Tax Calculator

Source code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Income Tax Calculator</title>

<script>

function calculateTax() {

    let name = prompt("Enter your full name:");

    if (!name || name.trim() === "") {

        alert("Name cannot be empty!");
```

```
    return;  
}
```

```
let pan = prompt("Enter your PAN number (10 characters, uppercase:");  
if (!pan || pan.length !== 10 || pan !== pan.toUpperCase()) {  
    alert("Invalid PAN format! It must be 10 characters and uppercase.");  
    return;  
}
```

```
let dobInput = prompt("Enter your Date of Birth (YYYY-MM-DD):");  
let dob = new Date(dobInput);  
if (isNaN(dob.getTime())) {  
    alert("Invalid Date format!");  
    return;  
}
```

```
let today = new Date();  
let age = today.getFullYear() - dob.getFullYear();  
if (  
    today.getMonth() < dob.getMonth() ||  
    (today.getMonth() === dob.getMonth() && today.getDate() < dob.getDate())  
) {  
    age--;
```

```
}
```

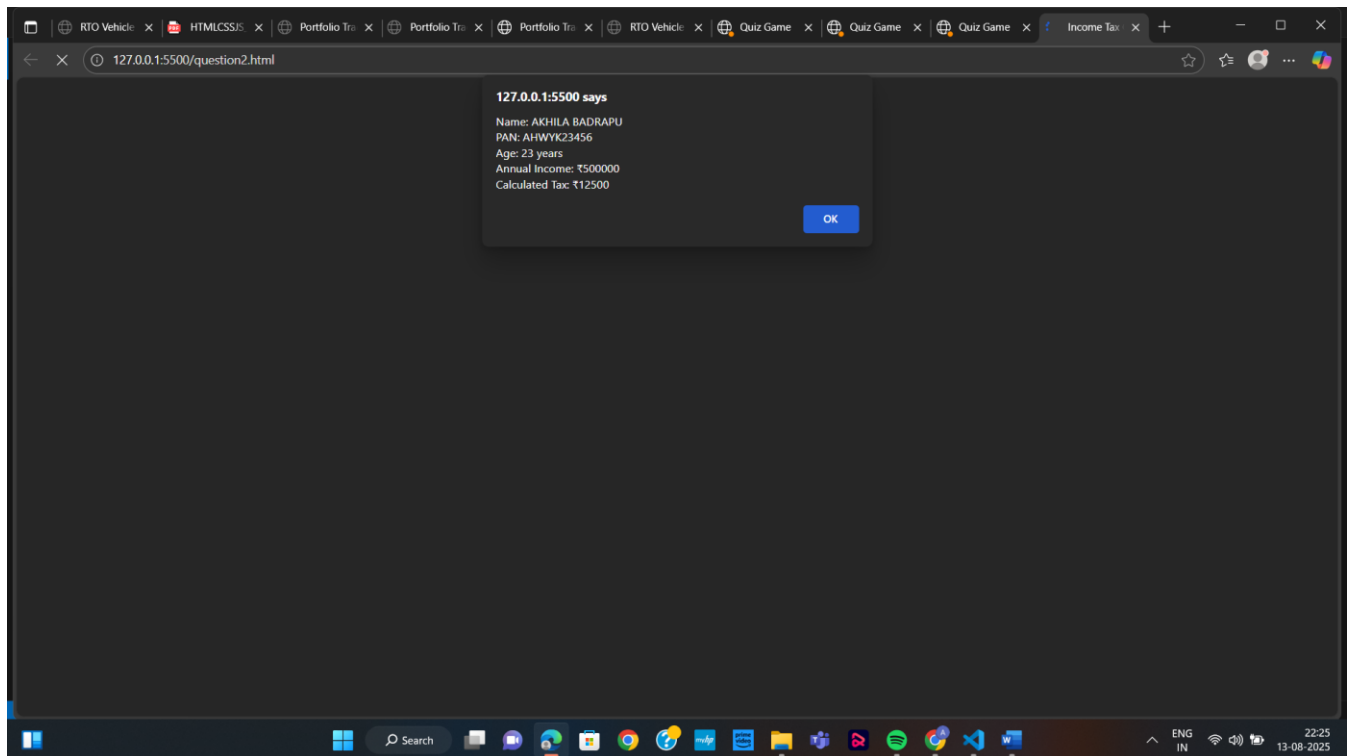
```
let income = parseFloat(prompt("Enter your annual income (in ₹):"));  
if (isNaN(income) || income < 0) {  
    alert("Invalid income amount!");  
    return;  
}
```

```
let tax = 0;  
if (income <= 250000) {  
    tax = 0;  
} else if (income <= 500000) {  
    tax = (income - 250000) * 0.05;  
} else if (income <= 1000000) {  
    tax = (250000 * 0.05) + ((income - 500000) * 0.2);  
} else {  
    tax = (250000 * 0.05) + (500000 * 0.2) + ((income - 1000000) * 0.3);  
}
```

```
tax = Math.round(tax);
```

```
let result =  
    "Name: " + name.toUpperCase() + "\n" +  
    "PAN: " + pan + "\n" +  
    "Age: " + age + " years\n" +  
    "Annual Income: ₹" + income + "\n" +  
    "Calculated Tax: ₹" + tax;  
  
    alert(result);  
}  
</script>  
</head>  
<body onload="calculateTax()">  
<h1>Income Tax Calculator</h1>  
<p>Reload the page to calculate again.</p>  
</body>  
</html>
```

Output:



### 3.Lab Question: Equality & Inequality Checker.

Source Code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Equality & Inequality Checker</title>
```

```
<script>
```

```
function compareValues() {
```

```
    let inputOne = prompt("Enter the first value:");
```

```
    let inputTwo = prompt("Enter the second value:");
```

```
let typeOne = typeof inputOne;
```

```
let typeTwo = typeof inputTwo;
```

```
let numberOne = Number(inputOne);
```

```
let numberTwo = Number(inputTwo);
```

```
let looseEqual = (inputOne == inputTwo);
```

```
let strictEqual = (inputOne === inputTwo);
```

```
let looseNotEqual = (inputOne != inputTwo);
```

```
let strictNotEqual = (inputOne !== inputTwo);
```

```
let output =
```

```
  "First Value: " + inputOne + " (" + typeOne + ")\n" +
```

```
  "Second Value: " + inputTwo + " (" + typeTwo + ")\n\n" +
```

```
  "Using == : " + looseEqual + " → Equality ignores type\n" +
```

```
  "Using === : " + strictEqual + " → Strict equality checks type too\n" +
```

```
  "Using != : " + looseNotEqual + " → Inequality ignores type\n" +
```

```
  "Using !== : " + strictNotEqual + " → Strict inequality checks type as well\n\n" +
```

```
  "Converted to Numbers for comparison:\n" +
```

```
  numberOne + " (" + typeof numberOne + "), " +
```

```
  numberTwo + " (" + typeof numberTwo + ")";
```

```
alert(output);
```

```
}
```

```
</script>
```

```
</head>
```

```
<body onload="compareValues()">
```

```
<h1>Equality & Inequality Checker</h1>
```

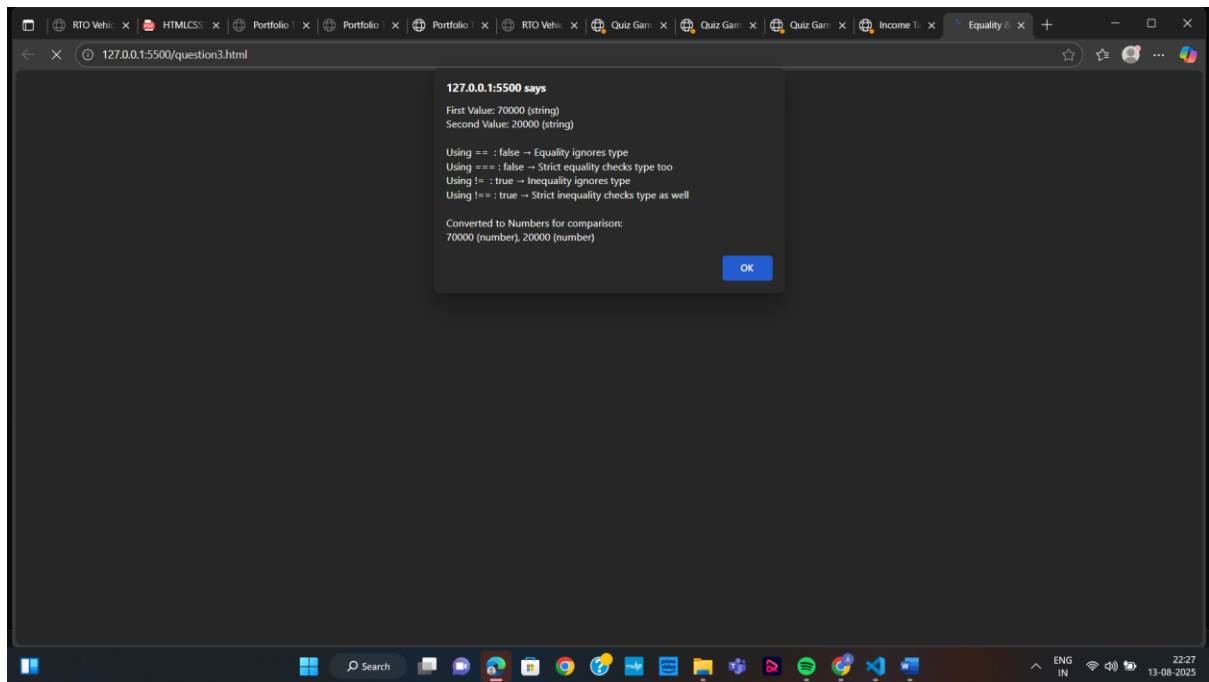


<p>Reload the page to compare again.</p>

</body>

</html>

Output:



#### 4. Lab Question: Bank Customers – Dynamic Array Creation & Manipulation

Source code:

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Bank Customers Management</title>

<script>

function manageCustomers() {

```
let totalCustomers = parseInt(prompt("How many customers to register?"));
if (isNaN(totalCustomers) || totalCustomers <= 0) {
    alert("Invalid number of customers!");
    return;
}
```

```
let customers = [];
for (let i = 0; i < totalCustomers; i++) {
    let name = prompt("Enter name of customer " + (i + 1) + ":");
    customers.push(name);
}
```

```
let namesList = "Customers using for...of:\n";
for (let cust of customers) {
    namesList += cust + "\n";
}
```

```
let idsList = "Customer IDs using for...in:\n";
for (let index in customers) {
    idsList += index + "\n";
}
```

```
customers.push("Anita");
```

```
let afterPush = "After push('Anita'): " + customers.join(", ");
```

```
customers.pop();
```

```
let afterPop = "After pop(): " + customers.join(", ");
```

```
customers.splice(1, 0, "Vikram");
```

```
let afterInsert = "After insert 'Vikram' at position 1: " + customers.join(", ");
```

```
customers.splice(2, 1);
```

```
let afterDelete = "After delete at position 2: " + customers.join(", ");
```

```
let finalList = "Final Customers: " + customers.join(", ");
```

```
alert(
```

```
  "Original Customers: " + customers.join(", ") + "\n\n" +
```

```
  namesList + "\n" +
```

```
  idsList + "\n" +
```

```
  afterPush + "\n" +
```

```
  afterPop + "\n" +
```

```
  afterInsert + "\n" +
```

```

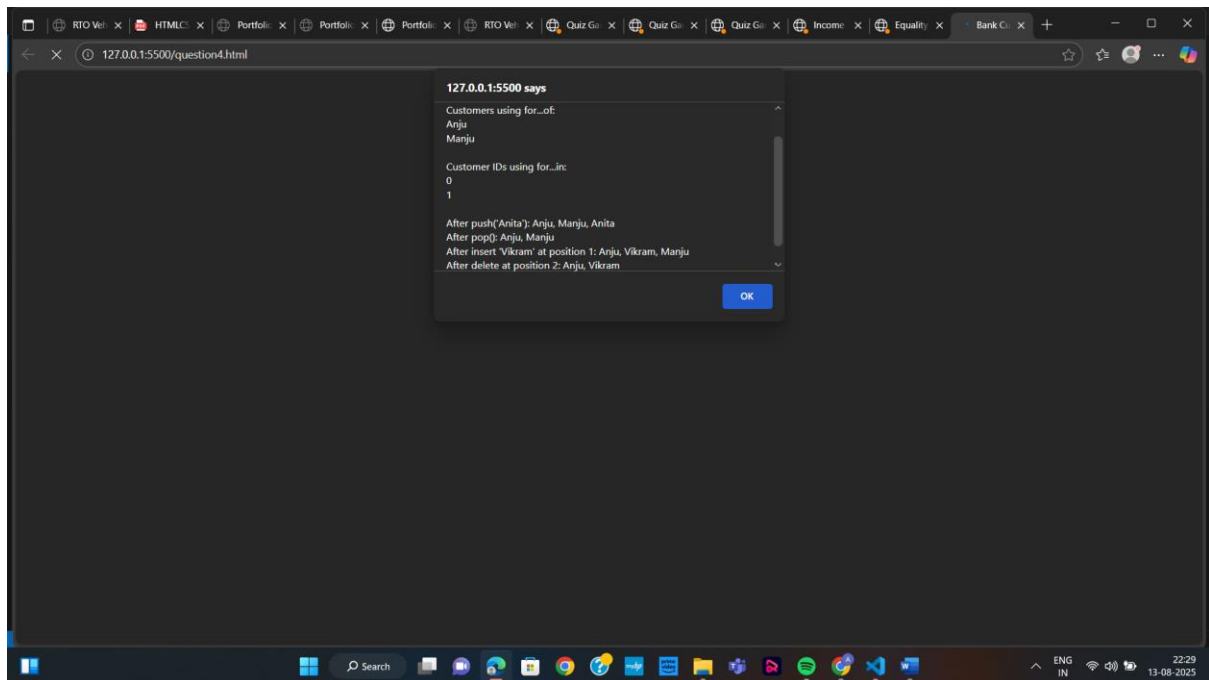
        afterDelete + "\n" +

        finalList

    );
}
</script>
</head>
<body onload="manageCustomers()">
<h1>Bank Customers – Dynamic Array Management</h1>
<p>Reload the page to run again.</p>
</body>
</html>

```

Output:



## 5. Lab Question: JSON – Book Store Management

Source code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Book Store Management</title>

<script>

function manageBooks() {

    let books = [

        { title: "HTML", author: "Akhila", price: 500, available: true },

        { title: "BootStrap", author: "Anjali", price: 450, available: false }

    ];

    let allBooks = "All Books:\n";

    for (let book of books) {

        allBooks += `Title: ${book.title}, Author: ${book.author}, Price: ₹${book.price}, Available:

        ${book.available}\n`;

    }

    let availableBooks = "Available Books:\n";

    for (let book of books) {
```

```
    if (book.available) {  
        availableBooks += `Title: ${book.title}, Author: ${book.author}, Price:  
₹${book.price}\n`;  
    }  
}
```

```
books.push({ title: "React.js", author: "Shankar", price: 600, available: true });  
let afterAdding = "After Adding:\n" + books.map(b => b.title).join(", ");
```

```
for (let book of books) {  
    if (book.title === "React") {  
        book.price = 500;  
    }  
}  
  
let afterPriceUpdate = "After Price Update:\n" + books.map(b => `${b.title} -  
₹${b.price}`).join("\n");
```

```
let indexToDelete = books.findIndex(b => b.title === "HTML");  
if (indexToDelete !== -1) {  
    books.splice(indexToDelete, 1);  
}  
  
let afterDeletion = "After Deletion:\n" + books.map(b => b.title).join(", ");
```

```

    alert(allBooks + "\n" + availableBooks + "\n" + afterAdding + "\n\n" + afterPriceUpdate +
"\n\n" + afterDeletion);
}

</script>

</head>

<body onload="manageBooks()">

<h1>Book Store Management</h1>

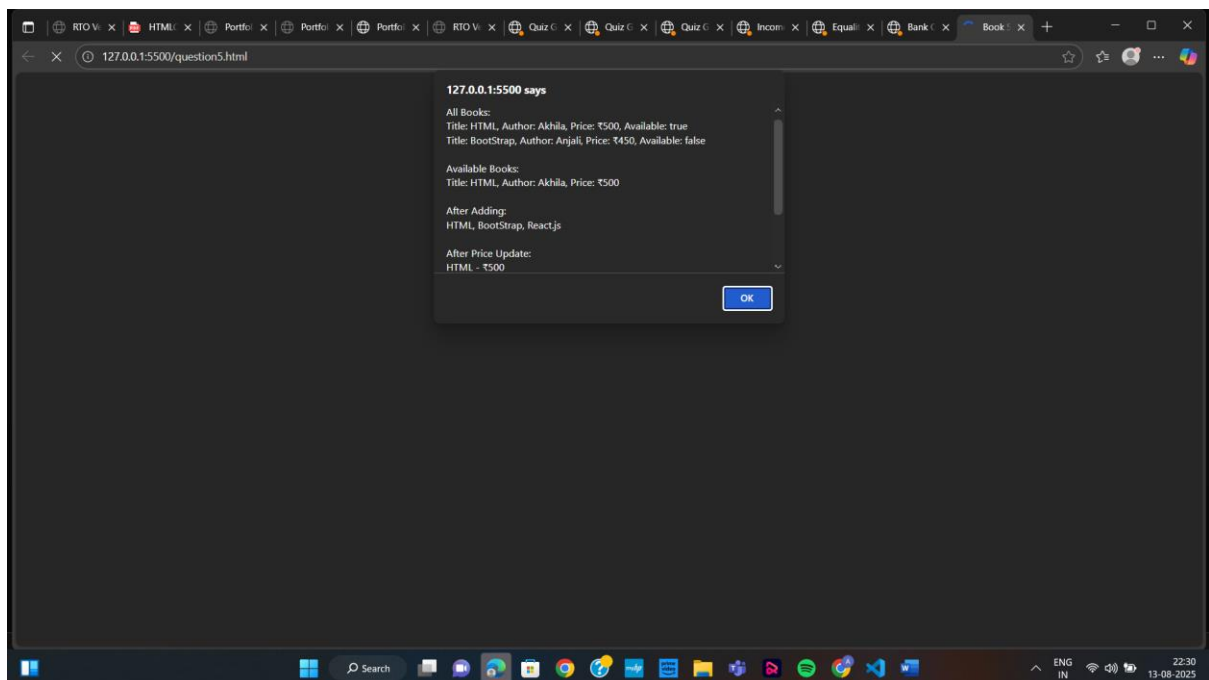
<p>Reload the page to run again.</p>

</body>

</html>

```

Output:



## 6. Lab Question: Tour Booking Form – getElementById()

Source Code:

```
<!DOCTYPE html>
```

```
<html lang="en">

<head>

<meta charset="UTF-8">

<title>Tour Booking Form</title>

<style>

  body {

    font-family: Arial, sans-serif;

    margin: 20px;

  }

  label {

    display: block;

    margin-top: 10px;

  }

  input, select, button {

    padding: 8px;

    margin-top: 5px;

    width: 250px;

  }

  #summary {

    margin-top: 20px;

    padding: 15px;

    border: 1px solid #ccc;

    background: #f9f9f9;

    width: 300px;

  }

</style>

<script>

function bookTour() {
```



```
let fullName = document.getElementById("name").value;
let email = document.getElementById("email").value;
let destination = document.getElementById("destination").value;
let days = parseInt(document.getElementById("days").value);
let travelers = parseInt(document.getElementById("travelers").value);
```

```
if (!fullName || !email || isNaN(days) || isNaN(travelers) || days <= 0 || travelers <= 0) {
    alert("Please fill in all fields correctly.");
    return;
}
```

```
let pricePerDay = 0;
if (destination === "Munnar") pricePerDay = 2000;
else if (destination === "Arunachala temple") pricePerDay = 3000;
else if (destination === "Wayand") pricePerDay = 4000;
```

```
let totalCost = pricePerDay * days * travelers;
```

```
let summary = `
    Name: ${fullName} <br>
```

```
Email: ${email} <br>
Destination: ${destination} <br>
Days: ${days} <br>
Travelers: ${travelers} <br>
Total Price: ₹${totalCost}
`;
document.getElementById("summary").innerHTML = summary;
}
</script>
</head>
<body>

<h2>Tour Booking Form</h2>
<form onsubmit="event.preventDefault(); bookTour();">
  <label>Full Name:</label>
  <input type="text" id="name">

  <label>Email:</label>
  <input type="email" id="email">

  <label>Destination:</label>
  <select id="destination">
    <option value="Munnar">Munnar</option>
    <option value="Arunachala temple">Arunachala temple</option>
    <option value="Wayand">Wayan</option>
  </select>

  <label>Number of Days:</label>
```

```
<input type="number" id="days">
```

```
<label>Number of Travelers:</label>
```

```
<input type="number" id="travelers">
```

```
<br><br>
```

```
<button type="submit">Book Now</button>
```

```
</form>
```

```
<div id="summary"></div>
```

```
</body>
```

```
</html>
```

Output:

The screenshot shows a web browser window with the address bar displaying '127.0.0.1:5500/question6.html'. The page contains a 'Tour Booking Form' with the following elements:

- Full Name:** A text input field containing 'Akhila'.
- Email:** A text input field containing 'aa@gmail.com'.
- Destination:** A dropdown menu with 'Munnar' selected.
- Number of Days:** A text input field containing '3'.
- Number of Travelers:** A text input field containing '3'.
- Book Now:** A button.
- Summary:** A box displaying the following information:
  - Name: Akhila
  - Email: aa@gmail.com
  - Destination: Munnar
  - Days: 3
  - Travelers: 3
  - Total Price: ₹18000

## 7. Lab Question – Sports Team Player Details

Output:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Sports Team Player Details</title>
<script>
function showPlayers() {
```

```
    const player1 = {
        name: "Virat Kohli",
        age: 36,
        sport: "Cricket",
        stats: {
            matches: 500,
            runs: 25000,
            average: 57.2
        }
    };
};
```

```
    const player2 = {
        name: "Shubman Gill",
        age: 38,
        sport: "Cricket",
        stats: {
            matches: 200,
```

```
        runs: 10000,
        average: 45
    }
};

//Store in array
const players = [player1, player2];

//Array destructuring
const [p1, p2] = players;

//Destructure Player 1 (Cricket)
const { name: name1, sport: sport1, stats: { matches: matches1, runs, average } } = p1;

//Destructure Player 2 (Cricket)
const { name: name2, sport: sport2, stats: { matches: matches2, goals, assists } } = p2;

// Output
let output = `
    Player ${name1} plays ${sport1}.
    Matches: ${matches1} | Runs: ${runs} | Batting Average: ${average}
    <br><br>
    Player ${name2} plays ${sport2}.
    Matches: ${matches2} | Goals: ${goals} | Assists: ${assists}
`;

// Display on page
document.getElementById("result").innerHTML = output;
```

```

}

</script>

</head>

<body onload="showPlayers()">

<h2>Sports Team Player Details</h2>

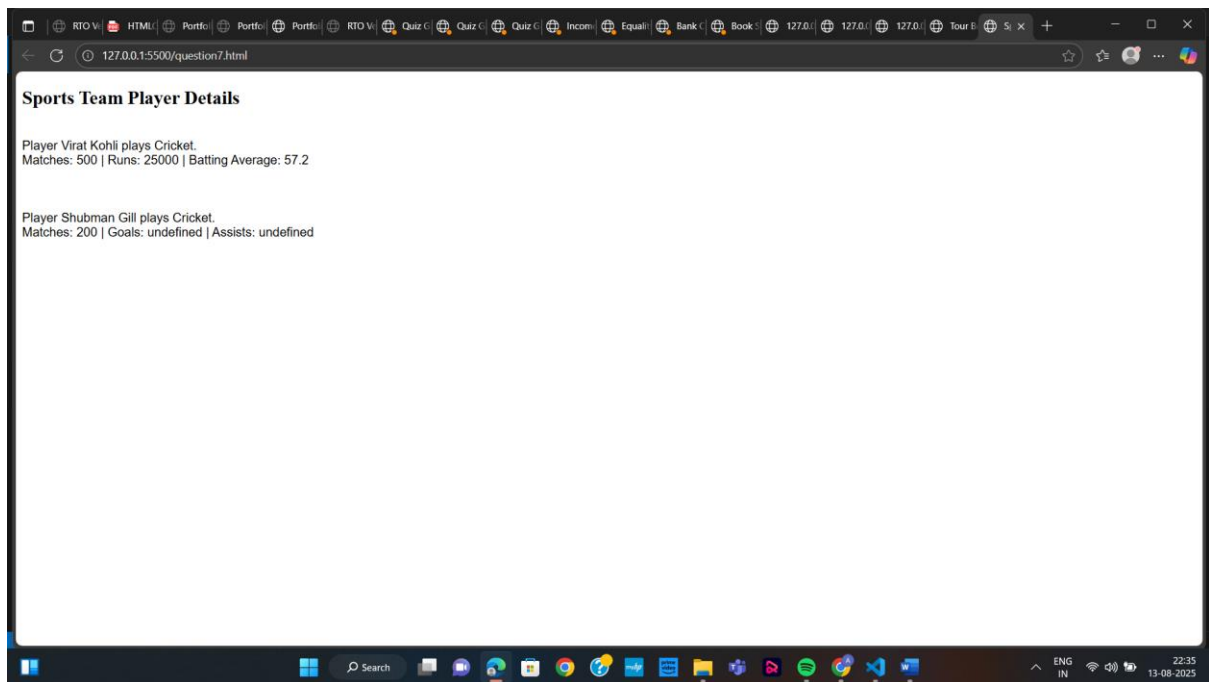
<div id="result" style="white-space: pre-line; font-family: Arial;"></div>

</body>

</html>

```

Output:



## 8.Lab Question – Switchboard Control (let, var, const)

Source Code:

```

<!DOCTYPE html>

<html lang="en">

```

```
<head>

<meta charset="UTF-8">

<title>Switchboard Control</title>

<script>

function controlSwitchboard() {

    const buildingName = "Sunrise Apartments";

    let status = "OFF";

    const devices = ["Light", "Fan", "AC"];

    for (var i = 0; i < devices.length; i++) {

        status = (status === "OFF") ? "ON" : "OFF";

        console.log(`Checking Switch ${i + 1}: ${devices[i]} is ${status}`);

    }

    console.log(`All switches checked for ${buildingName}.`);

    console.log(`Outside loop: var counter is ${i}`);

    try {

        buildingName = "Moonlight Residency";

    } catch (err) {

        console.error("Error when trying to change building name:", err.message);

    }

}
```

```

</script>

</head>

<body onload="controlSwitchboard()">

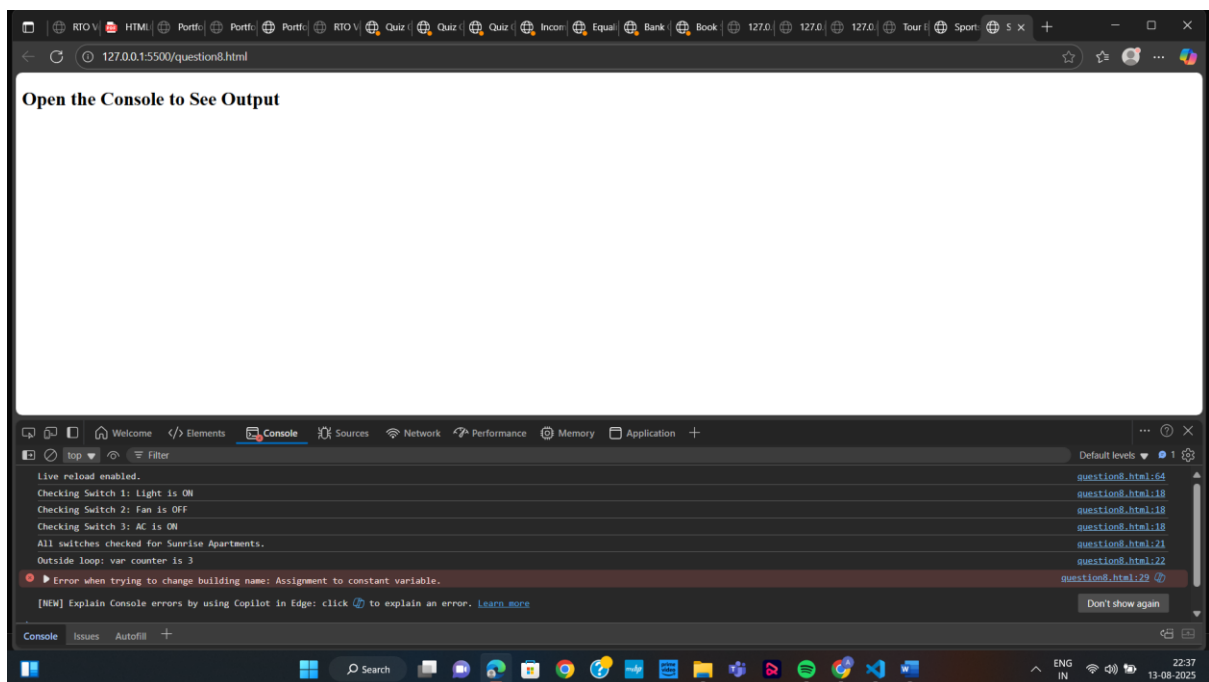
<h2>Open the Console to See Output</h2>

</body>

</html>

```

Output:



## 9. .Lab Question: Income Tax Calculator with Multiple Callbacks

Source Code:

```

<!DOCTYPE html>

<html>

<head>

    <title>Income Tax Calculator (Multiple Callbacks)</title>

</head>

```



```
<body>

  <h1>Income Tax Calculator (Multiple Callbacks)</h1>

  <div id="result"></div>

</script>

let taxRecords = [];

function calculateTax(name, pan, income, callback) {

  let tax = 0;

  if (income <= 250000) {

    tax = 0;

  } else if (income <= 500000) {

    tax = income * 0.05;

  } else if (income <= 1000000) {

    tax = income * 0.20;

  } else {

    tax = income * 0.30;

  }

  tax = Math.round(tax);

  callback(name, pan, income, tax);

}
```

```
function logToConsole(name, pan, income, tax) {  
    console.log(`Name: ${name}`);  
    console.log(`PAN: ${pan}`);  
    console.log(`Annual Income: ₹${income}`);  
    console.log(`Calculated Tax: ₹${tax}`);  
}
```

```
function displayOnPage(name, pan, income, tax) {  
    document.getElementById("result").innerHTML = `  
        <p><strong>Name:</strong> ${name}</p>  
        <p><strong>PAN:</strong> ${pan}</p>  
        <p><strong>Annual Income:</strong> ₹${income}</p>  
        <p><strong>Calculated Tax:</strong> ₹${tax}</p>  
    `;  
}
```

```
function saveToRecords(name, pan, income, tax) {  
    taxRecords.push({ name, pan, income, tax });  
    console.log("Saved Records:", taxRecords);  
}
```

```
let name = prompt("Enter Name:");
```

```
let pan = prompt("Enter PAN:");

let income = Number(prompt("Enter Annual Income:"));

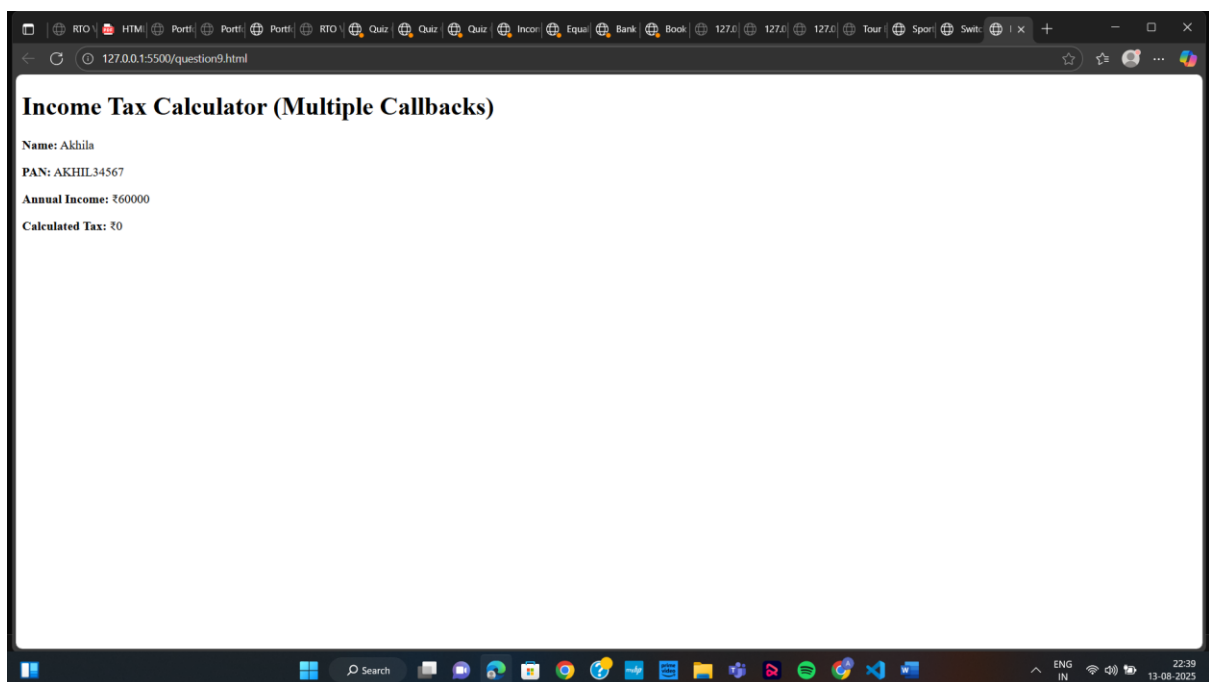

calculateTax(name, pan, income, logToConsole);

calculateTax(name, pan, income, displayOnPage);

calculateTax(name, pan, income, saveToRecords);

</script>
</body>
</html>
```

Output:



10. Build a multi-section RTO Vehicle Registration form with Bootstrap that validates:

Source Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>RTO Vehicle Registration</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

<style>

    .progress {

        height: 20px;

    }

</style>

</head>

<body class="bg-light">

<div class="container mt-4 mb-5">

    <h2 class="mb-4 text-center">RTO Vehicle Registration Form</h2>

    <!-- Progress Bar -->

    <div class="progress mb-4">

        <div id="formProgress" class="progress-bar bg-success" role="progressbar"
style="width: 0%">0%</div>

    </div>
```

```
<form id="rtoForm" class="needs-validation" novalidate>
```

```
<!-- Accordion -->
```

```
<div class="accordion" id="formAccordion">
```

```
<!-- Section 1: Owner -->
```

```
<div class="accordion-item">
```

```
<h2 class="accordion-header">
```

```
<button class="accordion-button" type="button" data-bs-toggle="collapse" data-  
bs-target="#ownerSection" aria-expanded="true">
```

```
1. Owner Details
```

```
</button>
```

```
</h2>
```

```
<div id="ownerSection" class="accordion-collapse collapse show" data-bs-  
parent="#formAccordion">
```

```
<div class="accordion-body row g-3">
```

```
<div class="col-md-6">
```

```
<label class="form-label">Full Name</label>
```

```
<input type="text" class="form-control" name="fullName" placeholder="e.g.  
Priya Natraajan" required>
```

```
<div class="invalid-feedback">Please enter full name.</div>
```

```
</div>
```

```
<div class="col-md-6">
```

```
<label class="form-label">Date of Birth</label>
```

```
<input type="date" class="form-control" name="dob" required>
```

```
<div class="invalid-feedback">You must be at least 18 years old.</div>
```

```
</div>
```

```
<div class="col-md-6">
```

```
<label class="form-label">Email</label>

<input type="email" class="form-control" name="email"
placeholder="name@example.com" required>

<div class="invalid-feedback">Enter a valid email.</div>
</div>

<div class="col-md-6">

<label class="form-label">Mobile (India)</label>

<input type="tel" class="form-control" name="mobile"
pattern="^\+91\d{10}$" placeholder="+91 98XXXXXXXX" required>

<div class="invalid-feedback">Enter a valid Indian mobile number
(+91XXXXXXXXXX).</div>

</div>

<div class="col-md-6">

<label class="form-label">Aadhaar</label>

<input type="text" class="form-control" name="aadhaar"
pattern="^\d{4}\s\d{4}\s\d{4}$" placeholder="1334 5679 9012" required>

<div class="invalid-feedback">Enter a valid 12-digit Aadhaar number.</div>

</div>

<div class="col-md-6">

<label class="form-label">PAN</label>

<input type="text" class="form-control" name="pan" pattern="^[A-Z]{5}[0-9]{4}[A-Z]{1}$" placeholder="ABCDE1234F" required>

<div class="invalid-feedback">Enter a valid PAN number.</div>

</div>

<div class="col-md-6">

<label class="form-label">Driving License No.</label>

<input type="text" class="form-control" name="dl" placeholder="Format:
State(2) + RTO(2) + 11 digits" required>

<div class="invalid-feedback">Enter a valid Driving License number.</div>

</div>
```

</div>

</div>

</div>

<!-- Section 2: Address -->

<div class="accordion-item">

<h2 class="accordion-header">

<button class="accordion-button collapsed" type="button" data-bs-toggle="collapse" data-bs-target="#addressSection">

2. Address

</button>

</h2>

<div id="addressSection" class="accordion-collapse collapse" data-bs-parent="#formAccordion">

<div class="accordion-body row g-3">

<div class="col-md-12">

<label class="form-label">Address Line 1</label>

<input type="text" class="form-control" name="address1" required>

<div class="invalid-feedback">Enter address.</div>

</div>

<div class="col-md-6">

<label class="form-label">City</label>

<input type="text" class="form-control" name="city" required>

<div class="invalid-feedback">Enter city.</div>

</div>

<div class="col-md-6">

<label class="form-label">State</label>

<select class="form-select" name="state" required>

<option value="">Choose...</option>

```

        <option>Tamil Nadu</option>
        <option>Karnataka</option>
        <option>Maharashtra</option>
    </select>

    <div class="invalid-feedback">Select a state.</div>
</div>

<div class="col-md-6">

    <label class="form-label">PIN Code</label>

    <input type="text" class="form-control" name="pin" pattern="^\d{6}$"
required>

    <div class="invalid-feedback">Enter valid 6-digit PIN code.</div>

</div>

</div>

</div>

</div>

<!-- Section 3: Vehicle -->

<div class="accordion-item">

    <h2 class="accordion-header">

        <button class="accordion-button collapsed" type="button" data-bs-
toggle="collapse" data-bs-target="#vehicleSection">

            3. Vehicle Details

        </button>

    </h2>

    <div id="vehicleSection" class="accordion-collapse collapse" data-bs-
parent="#formAccordion">

        <div class="accordion-body row g-3">

            <div class="col-md-6">

                <label class="form-label">Registration No.</label>

```



```
<input type="text" class="form-control" name="regNo" required>
<div class="invalid-feedback">Enter registration number.</div>
</div>
<div class="col-md-6">
  <label class="form-label">Vehicle Class</label>
  <select class="form-select" name="vehicleClass" required>
    <option value="">Choose...</option>
    <option>Car</option>
    <option>Bike</option>
    <option>Truck</option>
  </select>
  <div class="invalid-feedback">Select vehicle class.</div>
</div>
<div class="col-md-6">
  <label class="form-label">Fuel Type</label>
  <select class="form-select" name="fuel" required>
    <option value="">Choose...</option>
    <option>Petrol</option>
    <option>Diesel</option>
    <option>Electric</option>
  </select>
  <div class="invalid-feedback">Select fuel type.</div>
</div>
<div class="col-md-6">
  <label class="form-label">Make</label>
  <input type="text" class="form-control" name="make" required>
  <div class="invalid-feedback">Enter make.</div>
</div>
```

```
<div class="col-md-6">

  <label class="form-label">Engine No.</label>

  <input type="text" class="form-control" name="engineNo" pattern="^[a-zA-Z0-9]{6,12}$" required>

  <div class="invalid-feedback">Enter valid engine number.</div>

</div>
```

```
<div class="col-md-6">

  <label class="form-label">Year of Manufacture</label>

  <input type="number" class="form-control" name="year" min="1900" max="2025" required>

  <div class="invalid-feedback">Enter valid year.</div>

</div>
```

```
<div class="col-md-6">

  <label class="form-label">Purchase Date</label>

  <input type="date" class="form-control" name="purchaseDate" required>

  <div class="invalid-feedback">Date cannot be in the future.</div>

</div>
```

```
<div class="col-md-6">

  <label class="form-label">Color</label>

  <input type="text" class="form-control" name="color" required>

  <div class="invalid-feedback">Enter color.</div>

</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<!-- Section 4: Compliance -->
```

```
<div class="accordion-item">
```

```
  <h2 class="accordion-header">
```

```
<button class="accordion-button collapsed" type="button" data-bs-toggle="collapse" data-bs-target="#complianceSection">
```

#### 4. Compliance & Insurance

```
</button>
```

```
</h2>
```

```
<div id="complianceSection" class="accordion-collapse collapse" data-bs-parent="#formAccordion">
```

```
<div class="accordion-body row g-3">
```

```
<div class="col-md-6">
```

```
<label class="form-label">Emission Norm</label>
```

```
<select class="form-select" name="emission" required>
```

```
<option value="">Choose...</option>
```

```
<option>BS-III</option>
```

```
<option>BS-IV</option>
```

```
<option>BS-VI</option>
```

```
</select>
```

```
<div class="invalid-feedback">Select emission norm.</div>
```

```
</div>
```

```
<div class="col-md-6">
```

```
<label class="form-label">Insurance Policy No.</label>
```

```
<input type="text" class="form-control" name="policyNo" required>
```

```
<div class="invalid-feedback">Enter policy number.</div>
```

```
</div>
```

```
<div class="col-md-6">
```

```
<label class="form-label">Insurer</label>
```

```
<input type="text" class="form-control" name="insurer" required>
```

```
<div class="invalid-feedback">Enter insurer name.</div>
```

```
</div>
```

```
<div class="col-md-6">
```

```

        <label class="form-label">Insurance Expiry</label>

        <input type="date" class="form-control" name="insuranceExpiry" required>

        <div class="invalid-feedback">Date must be today or later.</div>

    </div>

</div>

</div>

</div>

```

```

<!-- Section 5: Finance -->

<div class="accordion-item">

    <h2 class="accordion-header">

        <button class="accordion-button collapsed" type="button" data-bs-
toggle="collapse" data-bs-target="#financeSection">

            5. Finance (Hypothecation)

        </button>

    </h2>

    <div id="financeSection" class="accordion-collapse collapse" data-bs-
parent="#formAccordion">

        <div class="accordion-body row g-3">

            <div class="col-md-12">

                <div class="form-check">

                    <input type="checkbox" class="form-check-input" id="hypothecation"
name="hypothecation">

                        <label class="form-check-label">Vehicle is under hypothecation /
loan</label>

                    </div>

                </div>

            <div id="financeDetails" class="row g-3 mt-2 d-none">

                <div class="col-md-6">

```

```

        <label class="form-label">Bank/NBFC</label>

        <input type="text" class="form-control" name="bank">

    </div>

    <div class="col-md-6">

        <label class="form-label">Agreement Date</label>

        <input type="date" class="form-control" name="agreementDate">

    </div>

</div>

</div>

</div>

</div>

</div>

<!-- Section 6: Uploads -->

<div class="accordion-item">

    <h2 class="accordion-header">

        <button class="accordion-button collapsed" type="button" data-bs-
toggle="collapse" data-bs-target="#uploadSection">

            6. Uploads

        </button>

    </h2>

    <div id="uploadSection" class="accordion-collapse collapse" data-bs-
parent="#formAccordion">

        <div class="accordion-body row g-3">

            <div class="col-md-6">

                <label class="form-label">ID/Address Proof</label>

                <input type="file" class="form-control" name="idProof"
accept=".pdf,.jpg,.jpeg,.png" required>

                <div class="invalid-feedback">Upload valid file (PDF/JPG/PNG ≤ 2 MB).</div>

            </div>

```

```
<div class="col-md-6">

    <label class="form-label">Insurance Copy</label>

    <input type="file" class="form-control" name="insuranceCopy"
accept=".pdf,.jpg,.jpeg,.png" required>

    <div class="invalid-feedback">Upload valid file (PDF/JPG/PNG ≤ 2 MB).</div>

</div>

<div class="col-md-6">

    <label class="form-label">Vehicle Image (optional)</label>

    <input type="file" class="form-control" name="vehicleImage"
accept=".jpg,.jpeg,.png">

    </div>

</div>

</div>

</div>
```

```
</div>
```

```
<!-- Terms & CAPTCHA -->
```

```
<div class="form-check mt-3">

    <input type="checkbox" class="form-check-input" name="terms" required>

    <label class="form-check-label">I agree to the terms and conditions</label>

    <div class="invalid-feedback">You must agree before submitting.</div>

</div>
```

```
<div class="mt-3">

    <label>CAPTCHA: <span id="captchaQuestion"></span></label>

    <input type="number" class="form-control" name="captcha" required>

    <div class="invalid-feedback">Please solve the CAPTCHA.</div>

</div>
```

```
        <button type="submit" id="submitBtn" class="btn btn-primary mt-4"
disabled>Submit</button>
```

```
    </form>
```

```
</div>
```

```
<!-- Toast -->
```

```
<div class="position-fixed bottom-0 end-0 p-3" style="z-index: 11">
```

```
    <div id="successToast" class="toast align-items-center text-bg-success border-0"
role="alert">
```

```
        <div class="d-flex">
```

```
            <div class="toast-body">
```

```
                Application submitted successfully!
```

```
            </div>
```

```
            <button type="button" class="btn-close btn-close-white me-2 m-auto" data-bs-
dismiss="toast"></button>
```

```
        </div>
```

```
    </div>
```

```
</div>
```

```
<script
```

```
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></scrip
t>
```

```
<script>
```

```
document.addEventListener("DOMContentLoaded", function() {
```

```
    const form = document.getElementById("rtoForm");
```

```
    const progressBar = document.getElementById("formProgress");
```

```
    const submitBtn = document.getElementById("submitBtn");
```

```
const captchaQ = document.getElementById("captchaQuestion");
```

```
let captchaAnswer = 0;
```

```
function generateCaptcha() {
```

```
    const a = Math.floor(Math.random() * 10) + 1;
```

```
    const b = Math.floor(Math.random() * 10) + 1;
```

```
    captchaAnswer = a + b;
```

```
    captchaQ.textContent = `${a} + ${b} = ?`;
```

```
}
```

```
generateCaptcha();
```

```
// Show finance details when checkbox checked
```

```
document.getElementById("hypothecation").addEventListener("change", function() {
```

```
    document.getElementById("financeDetails").classList.toggle("d-none", !this.checked);
```

```
});
```

```
// Validation check
```

```
form.addEventListener("input", function() {
```

```
    let validSections = 0;
```

```
    document.querySelectorAll(".accordion-item").forEach(section => {
```

```
        if (section.querySelectorAll(":invalid").length === 0) {
```

```
            validSections++;
```

```
        }
```

```
    });
```

```
    const percent = Math.floor((validSections / document.querySelectorAll(".accordion-item").length) * 100);
```

```
    progressBar.style.width = percent + "%";
```



```

    progressBar.textContent = percent + "%";
    submitBtn.disabled = !form.checkValidity();
});

form.addEventListener("submit", function(e) {
    e.preventDefault();
    if (parseInt(form.captcha.value) !== captchaAnswer) {
        form.captcha.setCustomValidity("Invalid captcha");
        form.captcha.reportValidity();
        return;
    } else {
        form.captcha.setCustomValidity("");
    }

    if (!form.checkValidity()) {
        e.stopPropagation();
    } else {
        const toast = new bootstrap.Toast(document.getElementById("successToast"));
        toast.show();
        form.reset();
        generateCaptcha();
        progressBar.style.width = "0%";
        progressBar.textContent = "0%";
        submitBtn.disabled = true;
    }
    form.classList.add("was-validated");
});
});

```

</script>

</body>

</html>

Output:

### RTO Vehicle Registration Form

1. Owner Details

Full Name

e.g. Priya Natraajan

Date of Birth

dd-mm-yyyy

Email

name@example.com

Mobile (India)

+91 98XXXXXXX

Aadhaar

1334 5679 9012

PAN

ABCDE1234F

Driving License No.

Format: State(2) + RTO(2) + 11 digits

2. Address

3. Vehicle Details

4. Compliance & Insurance

5. Finance (Hypothecation)

6. Uploads

☐ I agree to the terms and conditions

CAPTCHA: 3 + 4 = ?

Submit

## 11. Lab Question – JavaScript Closure (Share Market Example)

Source Code:

Question11.js ->js file

```
function createPortfolioTracker() {  
    // Private portfolio array (accessible only via closure)  
    let portfolio = [];  
  
    // Buy shares  
    function buyShare(company, quantity, pricePerShare) {
```

```
portfolio.push({ company, quantity, pricePerShare });  
  
console.log(`Bought ${quantity} shares of ${company} at ₹${pricePerShare} each.`);  
}
```

```
// Sell shares
```

```
function sellShare(company, quantity) {  
  let found = false;  
  for (let share of portfolio) {  
    if (share.company === company) {  
      if (share.quantity >= quantity) {  
        share.quantity -= quantity;  
        console.log(`Sold ${quantity} shares of ${company}.`);  
      } else {  
        console.log(`Not enough shares to sell for ${company}.`);  
      }  
      found = true;  
      break;  
    }  
  }  
  if (!found) {  
    console.log(`No shares found for ${company}.`);  
  }  
}
```

```
// Calculate total portfolio value
```

```
function totalValue() {  
  let total = 0;  
  for (let share of portfolio) {
```

```

        total += share.quantity * share.pricePerShare;
    }

    console.log(`Portfolio Value: ₹${total}`);
}

// Public methods
return {
    buyShare,
    sellShare,
    totalValue
};
}

// --- Usage Example ---
const myPortfolio = createPortfolioTracker();

myPortfolio.buyShare("TCS", 10, 3500);
myPortfolio.buyShare("Infosys", 5, 1500);
myPortfolio.sellShare("TCS", 3);
myPortfolio.totalValue();

```

HTML FILE → index.html

```

<!DOCTYPE html>

<html>

<head>

    <title>Portfolio Tracker</title>

```

```
</head>

<body>

  <script src="question11.js"></script>

</body>

</html>
```

Output:

